

Introduction to version control with Git

Scientific workflows: Tools and Tips 

2025-07-17

Scientific workflows: Tools and Tips

 Every 3rd Thursday  4-5 p.m.  Webex

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Why version control?



Why version control?

Git is like a Lab Notebook for your scripts

- **Tracks** every change in your scripts
- Helps you **recover** older versions
- Enables safe and easy **collaboration**
- Makes it easy to **share** your work with others

Today

- Introduction to **Git**
- Simple Git workflow in **theory and practice**
- Publish your work on **GitHub**
- Find detailed how-to guides on the website

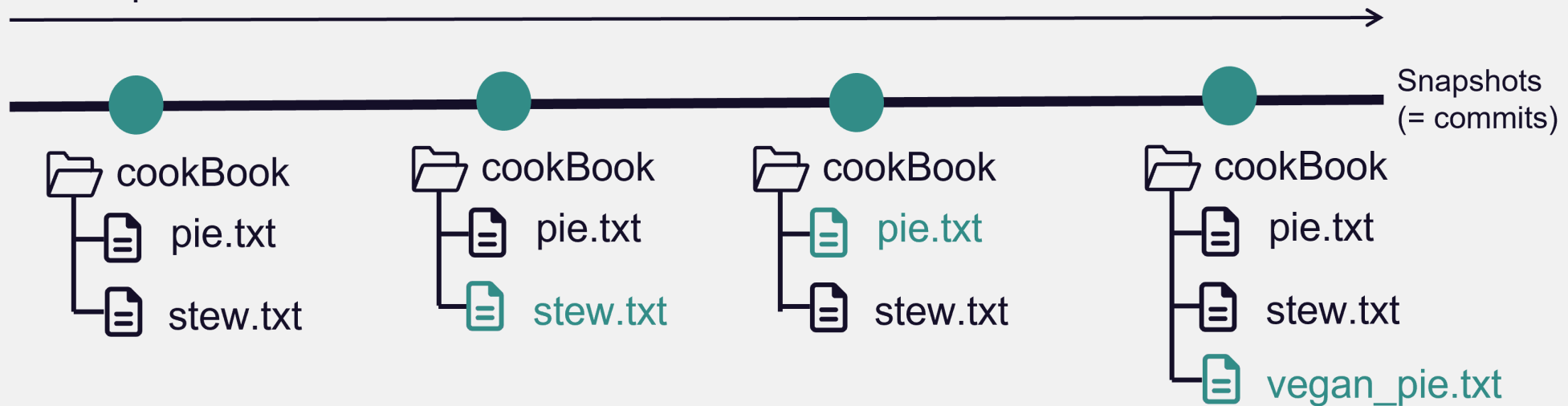
What is Git?

- **Open source and free** to use version control software
- Quasi **standard** for software development
- **Complete** and **long-term** history of every file in your project
- A whole universe of **other software and services** around it

What is Git?

- For projects with **mainly text files** (e.g. code, markdown files, ...)
- Basic idea: Take snapshots (**commits**) of your project over time

Development over time

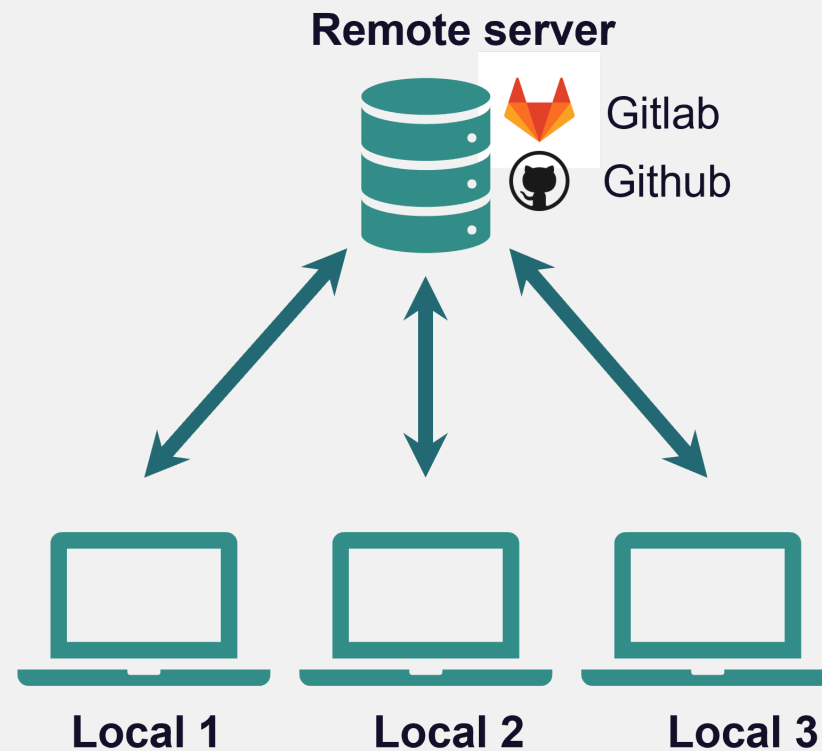


- A project version controlled with Git is a Git **repository (repo)**

Version control with Git

Git is a **distributed** version control system

Idea: many *local* repositories synced via one *remote* repo



How to use Git

After you **installed it** there are different ways to interact with the software.

How to use Git - Terminal

Using Git from the terminal

```
Selina_User@DESKTOP-G0RM7MS MINGW64 ~/Files_Selina
$ cd Repos/02_workshops/first_git_project/

Selina_User@DESKTOP-G0RM7MS MINGW64 ~/Files_Selina/Repos/02_workshops/first_git_
project
$ git init
Initialized empty Git repository in C:/Users/Selina_User/Files_Selina/Repos/02_w
orkshops/first_git_project/.git/

Selina_User@DESKTOP-G0RM7MS MINGW64 ~/Files_Selina/Repos/02_workshops/first_git_
project (master)
$
```

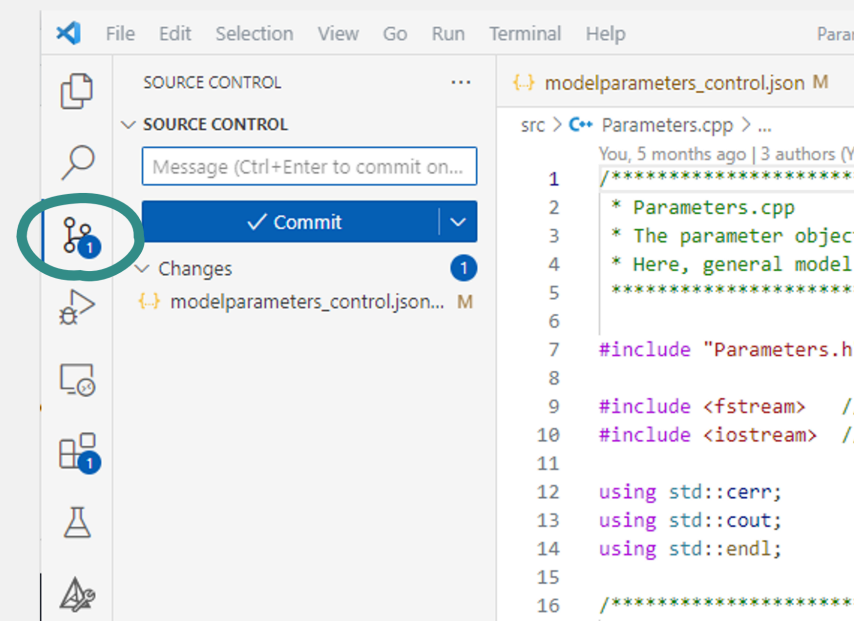
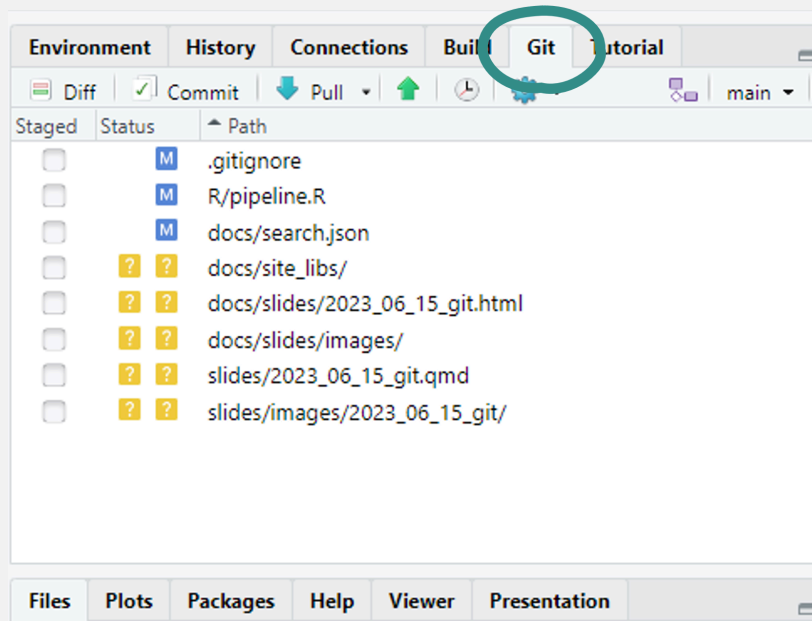
+ Most control

— You need to use terminal 🤖

+ A lot of help/answers online

How to use Git - Integrated GUIs

A Git GUI is integrated in most (all?) IDEs, e.g. R Studio, VS Code



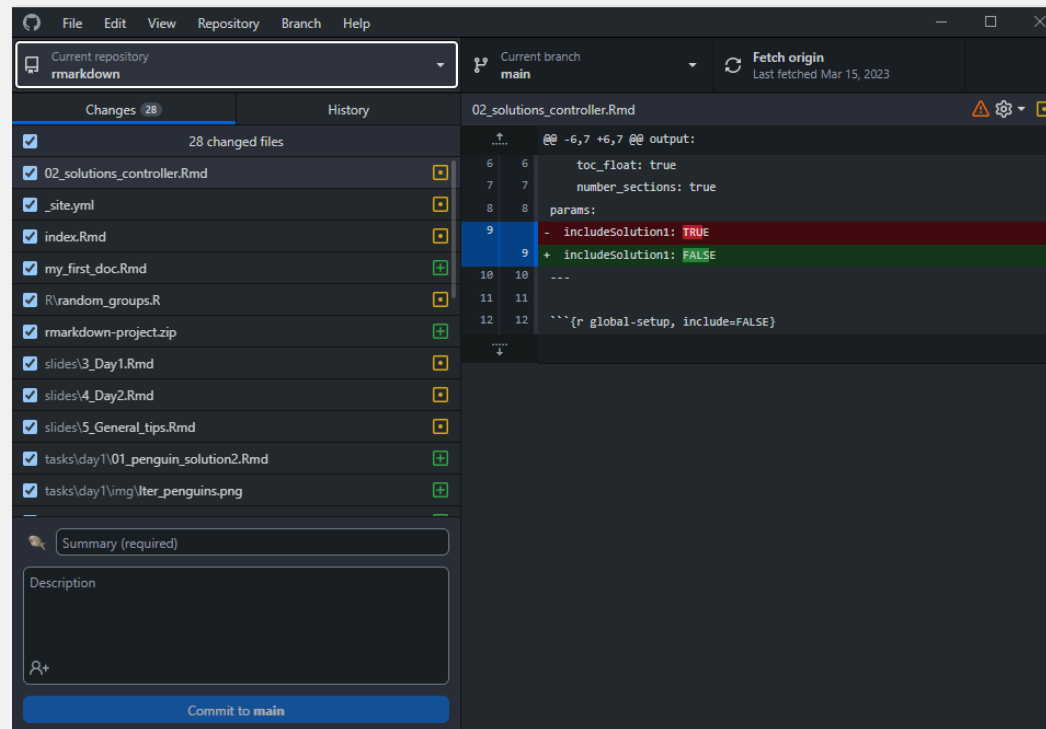
+ Easy and intuitive

+ Stay inside IDE

- Different for every program

How to use Git - Standalone GUIs

Standalone Git GUI software, e.g. GitHub Desktop, Source Tree, ...



- + Easy and intuitive
- + Use for all projects

— Switch programs to use Git

How to use Git

Which one to choose?

- Depends on experience and taste
- You can mix methods because they are all interfaces to the same Git
- We will use GitHub Desktop
 - Beginner-friendly, intuitive and convenient
 - Nice integration with GitHub



Tip

Have a look [here](#) to find **How-To guides for the other methods** as well.

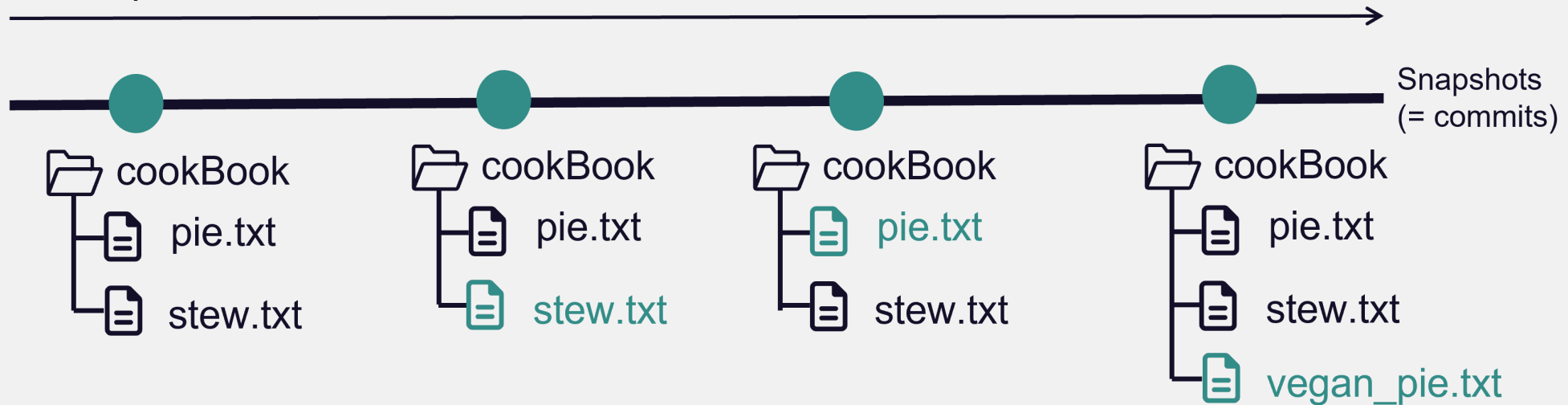
The basic Git workflow

```
git init, git add, git commit, git push
```

Example

A cook book project to collect all my favorite recipes.

Development over time



In real life this would be e.g. a data analysis project, your thesis in LaTeX, a software project, ...

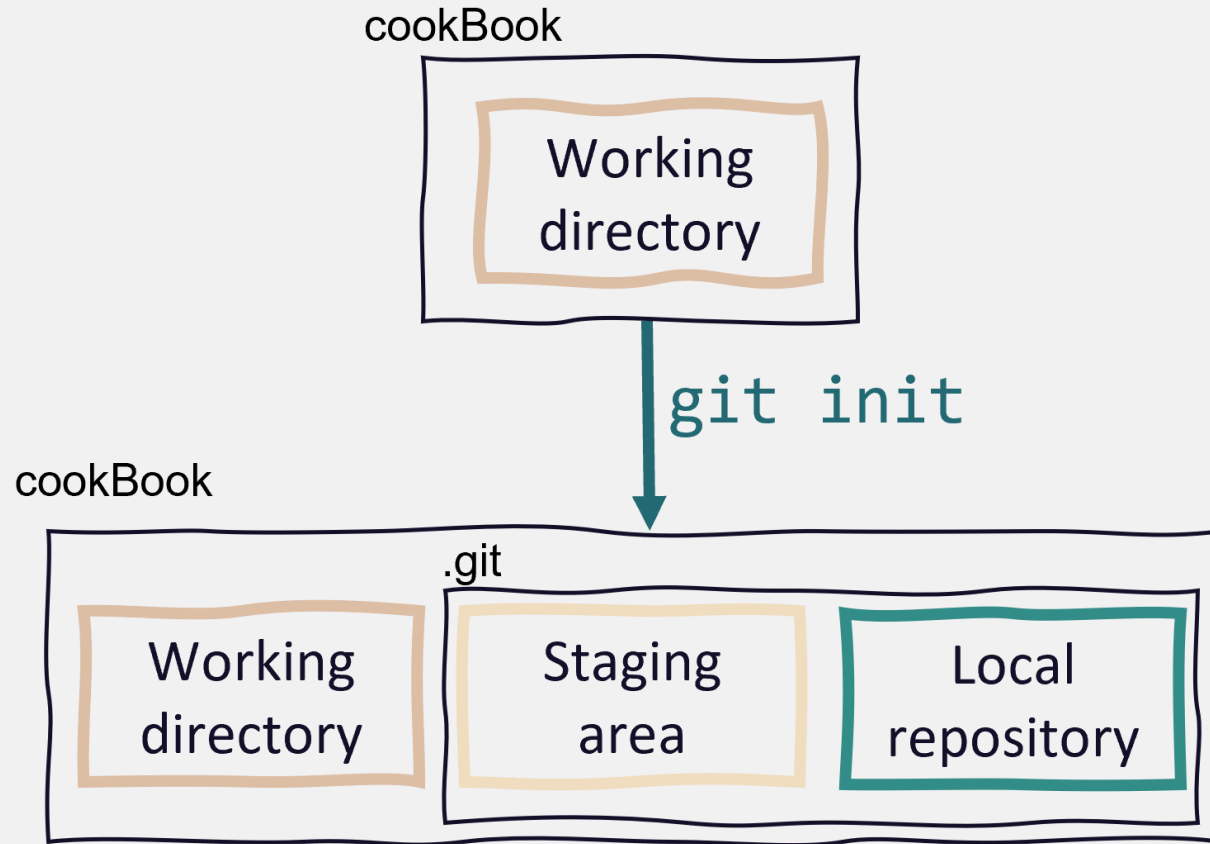
Step 1: Initialize a Git repository

cookBook



 cookBook

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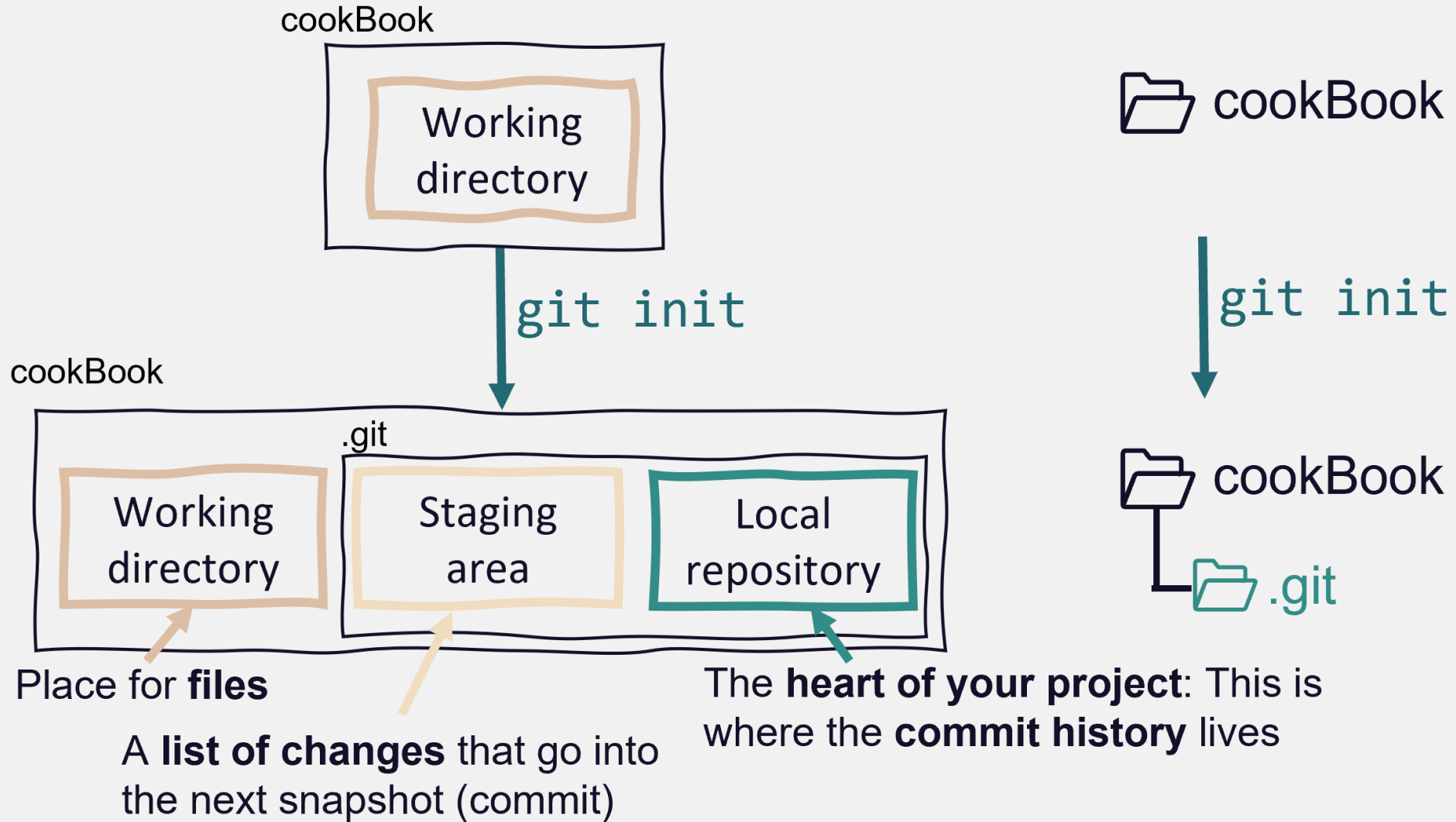


📁 cookBook

↓ git init

📁 cookBook
└── 📁 .git

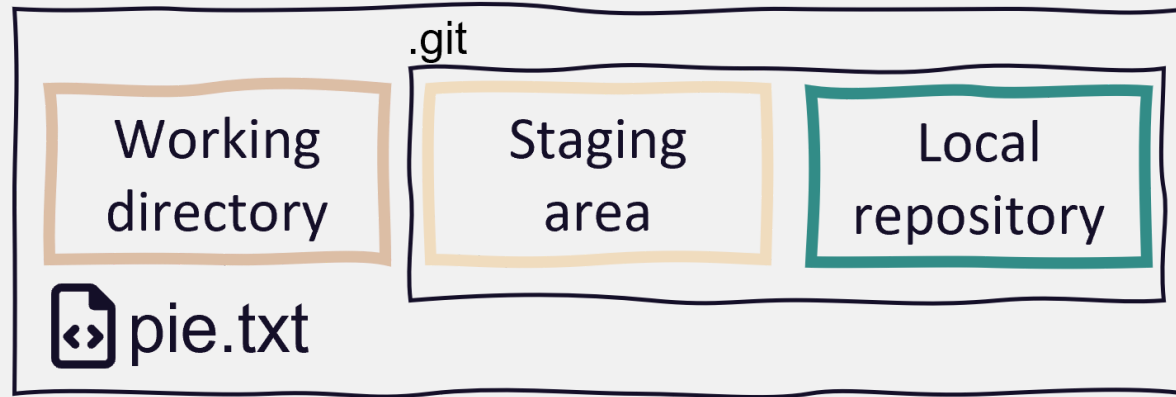
Step 1: Initialize a Git repository



Step 2: Add and modify files

Git detects any changes in the working directory

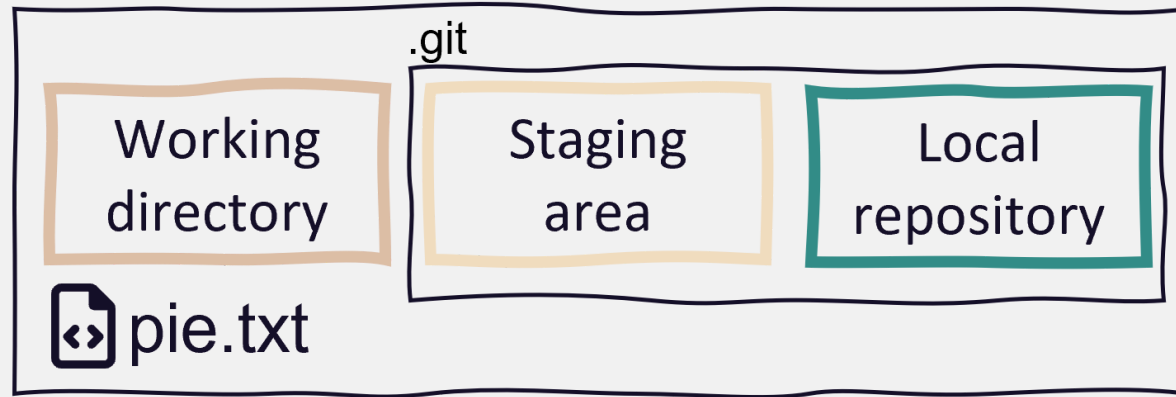
cookBook



Step 2: Stage changes

Staging a file means to **list it for the next commit**.

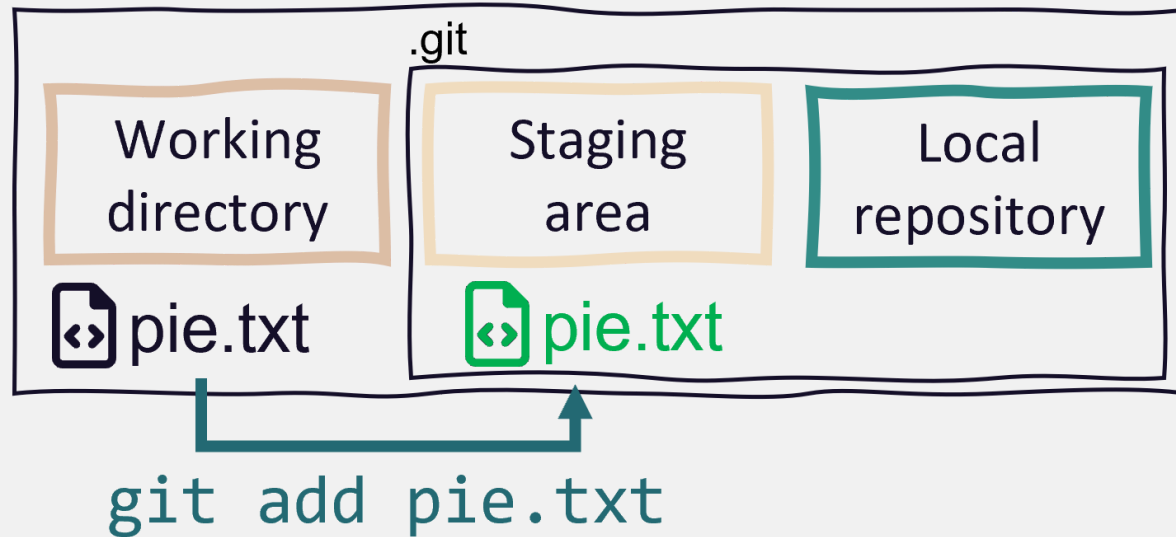
cookBook



Step 2: Stage changes

Staging a file means to **list it for the next commit**.

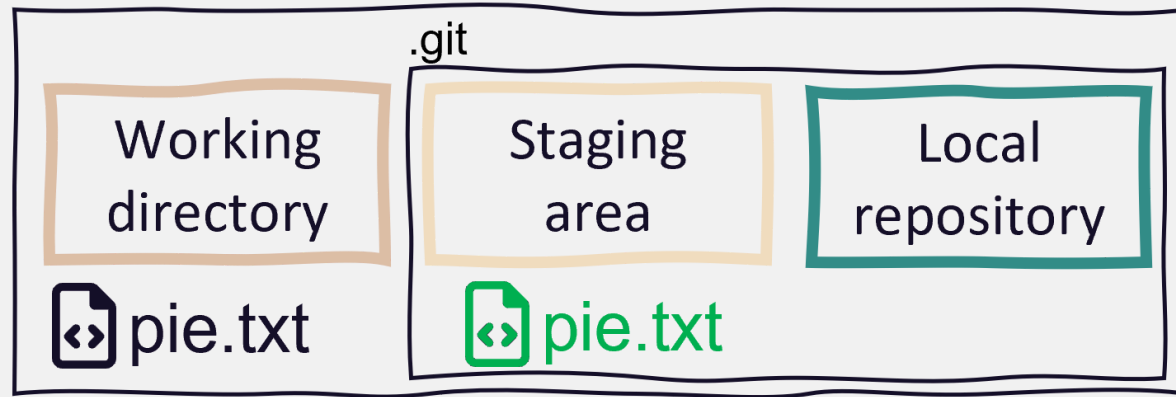
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Step 3: Commit changes

Commits are the snapshots of your project state

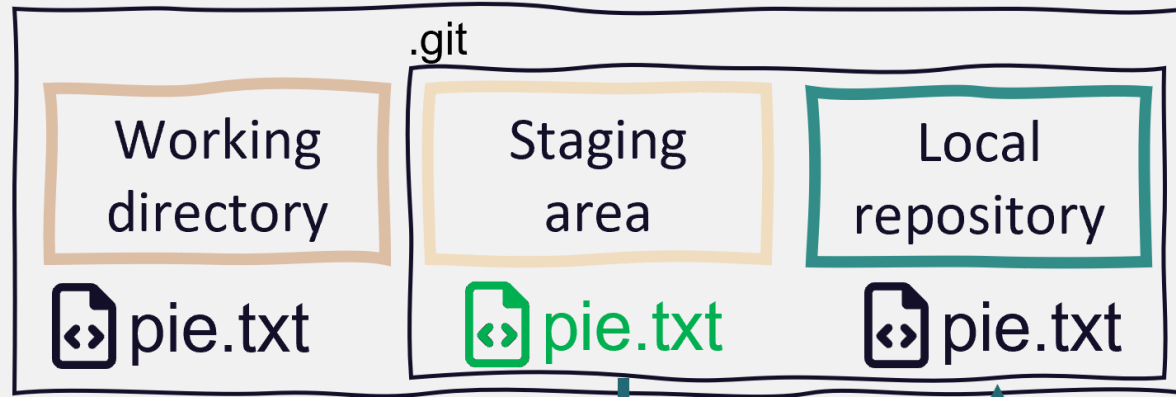
cookBook



Step 3: Commit changes

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cookBook



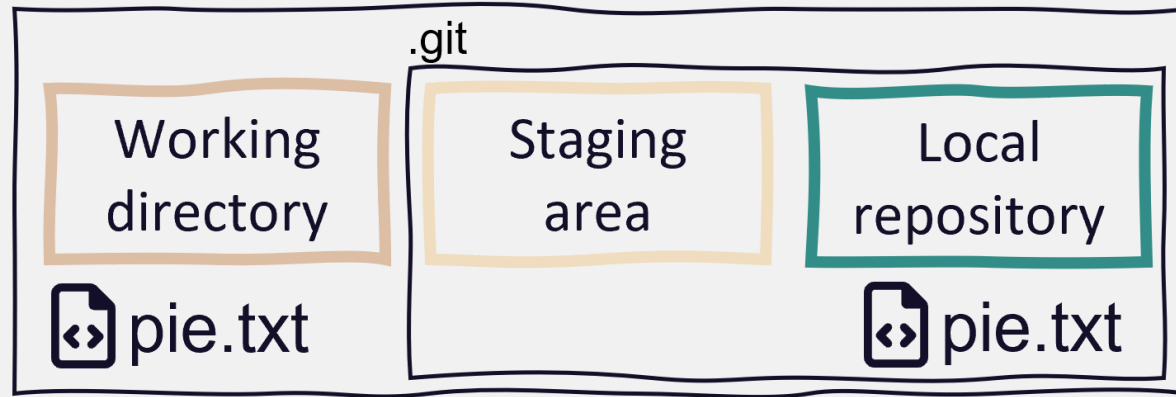
```
git commit -m „Add pie recipe“
```



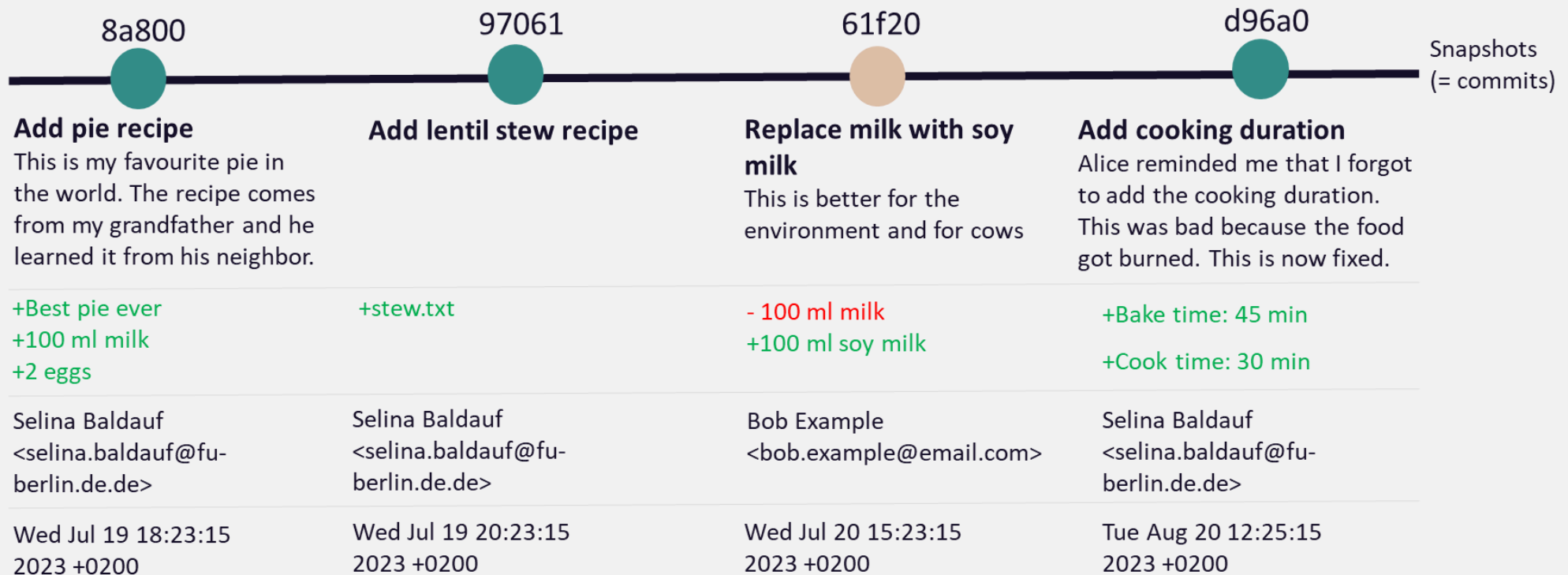
Step 3: Commit changes

Changes are part of Git history and staging area is clear again

cookBook



The commit history



Good commit messages

	COMMENT	DATE
○	CREATED MAIN LOOP & TIMING CONTROL	14 HOURS AGO
○	ENABLED CONFIG FILE PARSING	9 HOURS AGO
○	MISC BUGFIXES	5 HOURS AGO
○	CODE ADDITIONS/EDITS	4 HOURS AGO
○	MORE CODE	4 HOURS AGO
○	HERE HAVE CODE	4 HOURS AGO
○	AAAAAAAAA	3 HOURS AGO
○	ADKFJSLKDFJSDKLFJ	3 HOURS AGO
○	MY HANDS ARE TYPING WORDS	2 HOURS AGO
○	HAAAAAAAAAANDS	2 HOURS AGO

AS A PROJECT DRAGS ON, MY GIT COMMIT MESSAGES GET LESS AND LESS INFORMATIVE.

xkcd on commit messages

Good commit messages

Good commit messages are descriptive and helpful.



Add pie recipe

This is my favorite pie in the world.
The recipe comes from my grandfather and
he learned it from his neighbor.



added a file.

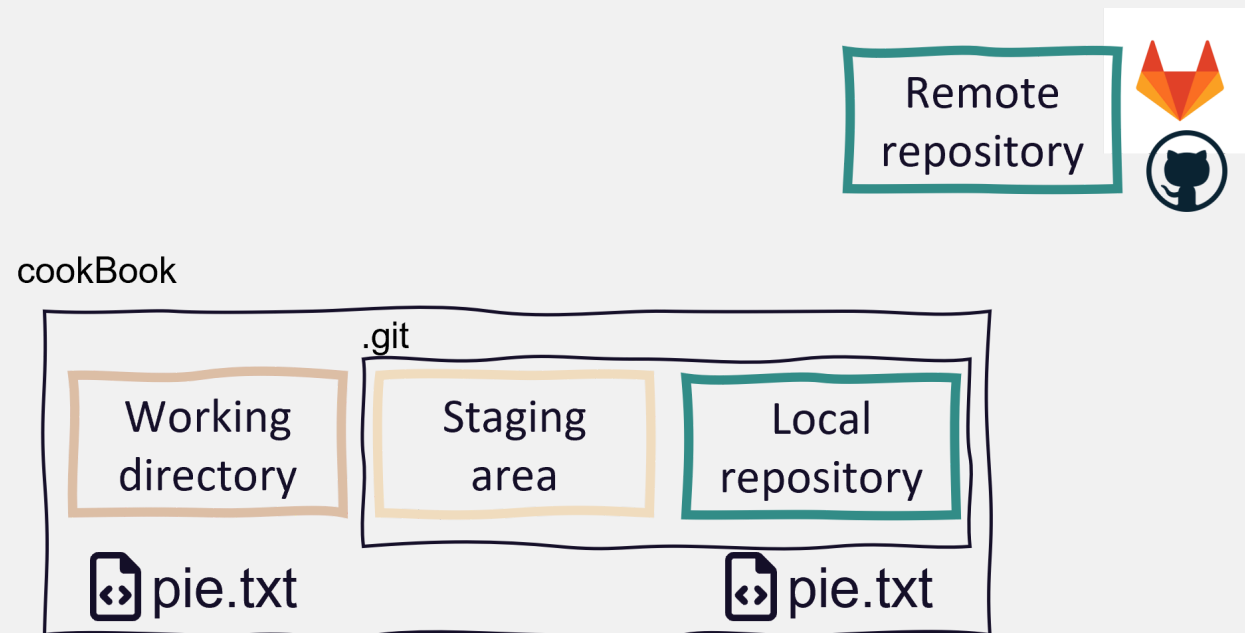
This is really good.

See [here](#) for more details on good commit messages.

Step 4: Share changes with the remote repo

Use remote repos (on a server) to **backup**, **synchronize**, **share** and **collaborate**

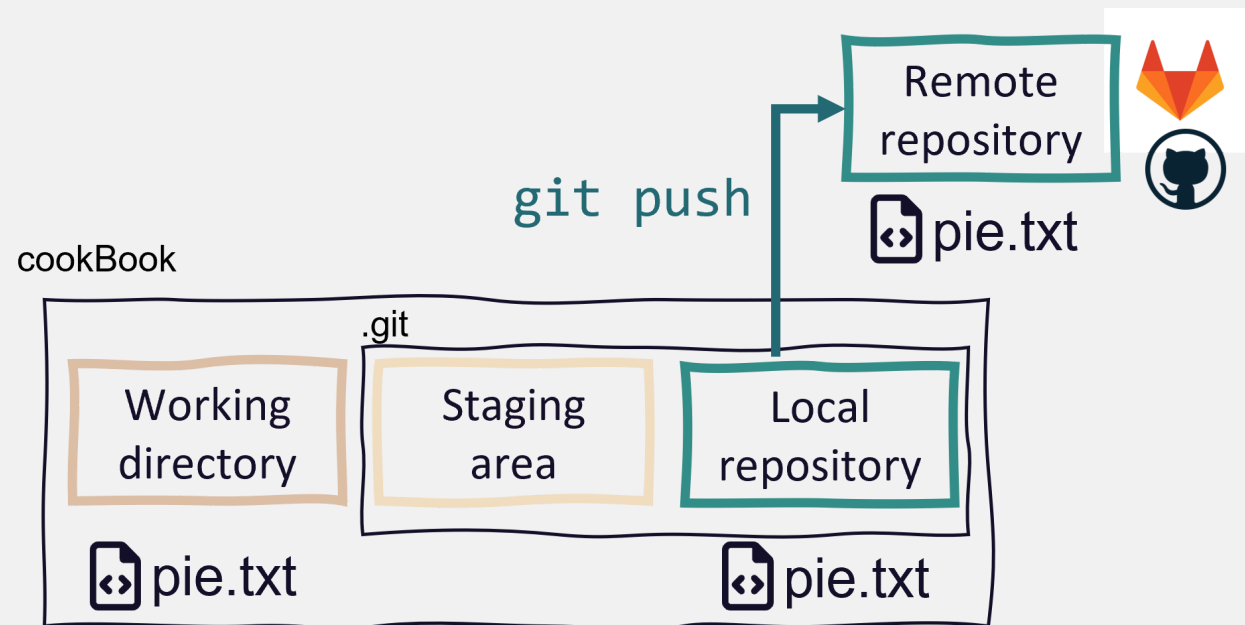
- can be **private** (you + collaborators) or **public** (visible to anyone)



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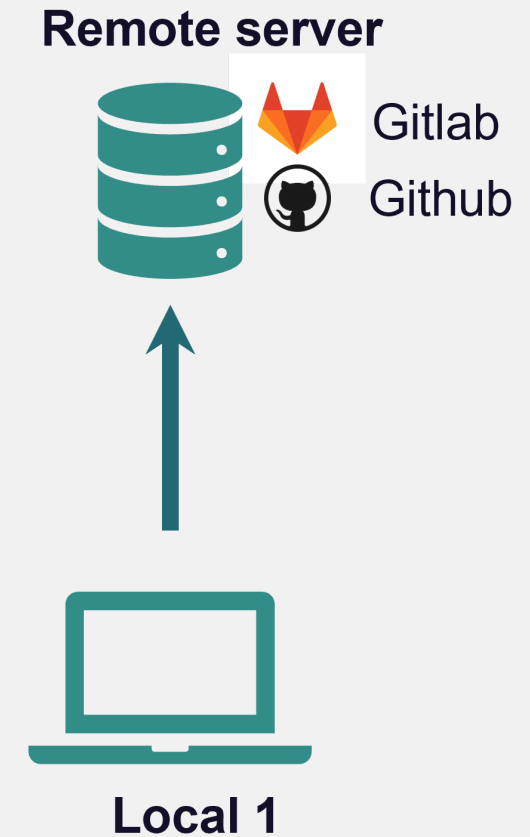
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Recap

Basic Git workflow:

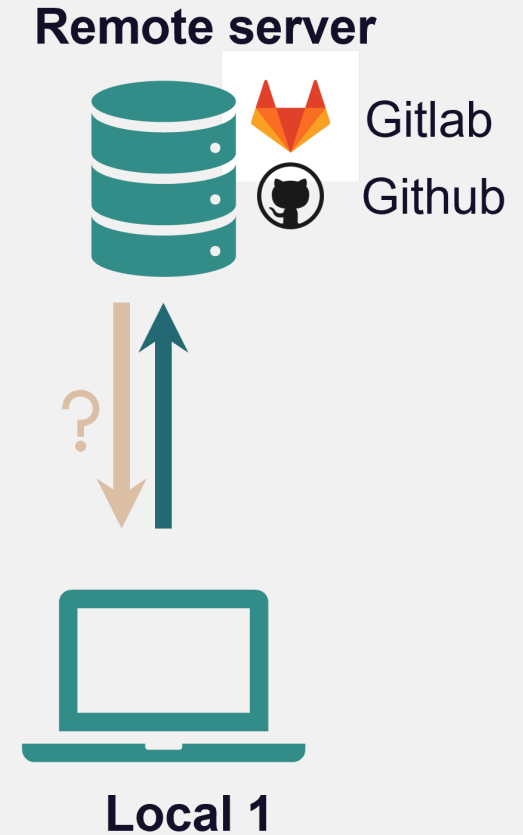
1. **Initialize** a Git repository
2. **Work** on the project
3. **Stage** and **commit** changes to the local repository
4. **Push** to the remote repository



Recap

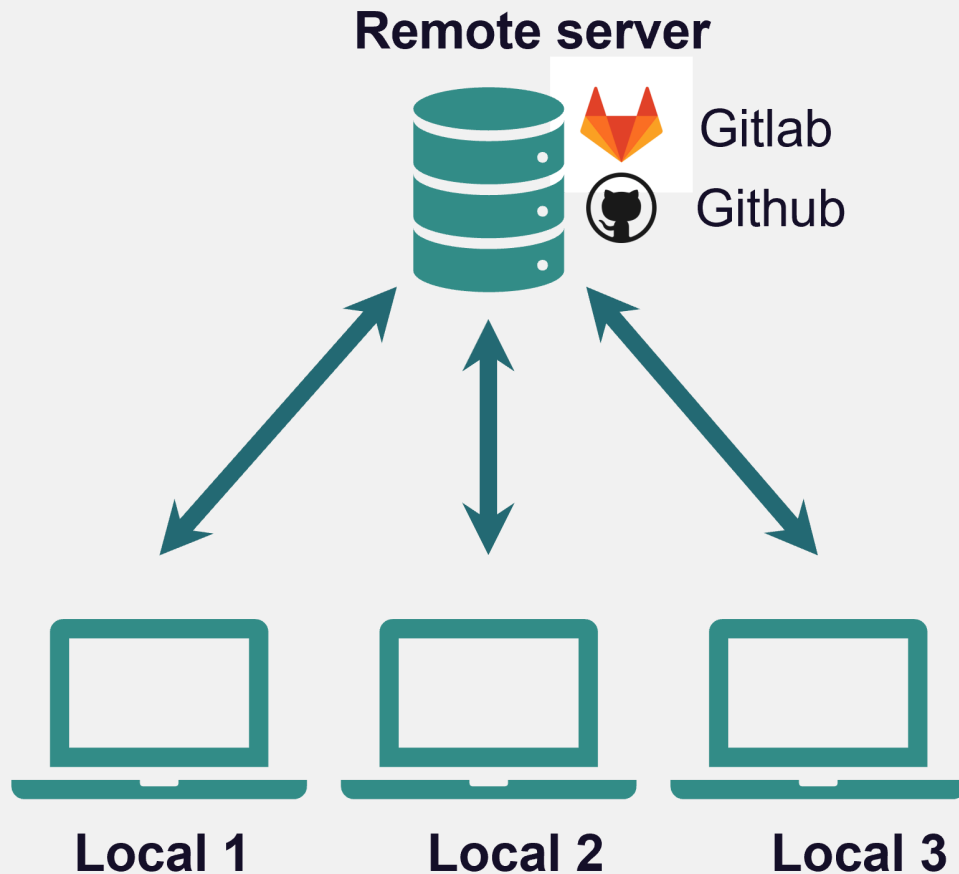
Basic Git workflow:

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4. **Push** to the remote repository



Recap

Git is a **distributed version control system**

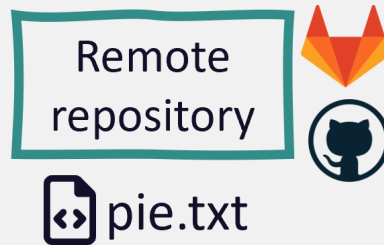


- Idea: many *local* repositories synced via one *remote* repo
- Collaborate with
 - **yourself** on different machines
 - your **colleagues** and friends
 - **strangers** on open source projects

Get a repo from a remote

In Git language, this is called **cloning**

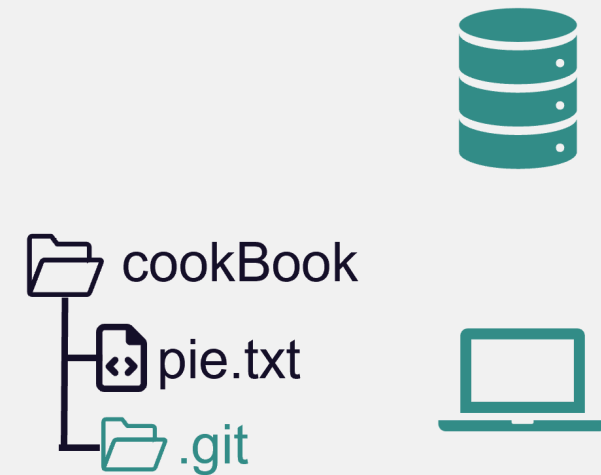
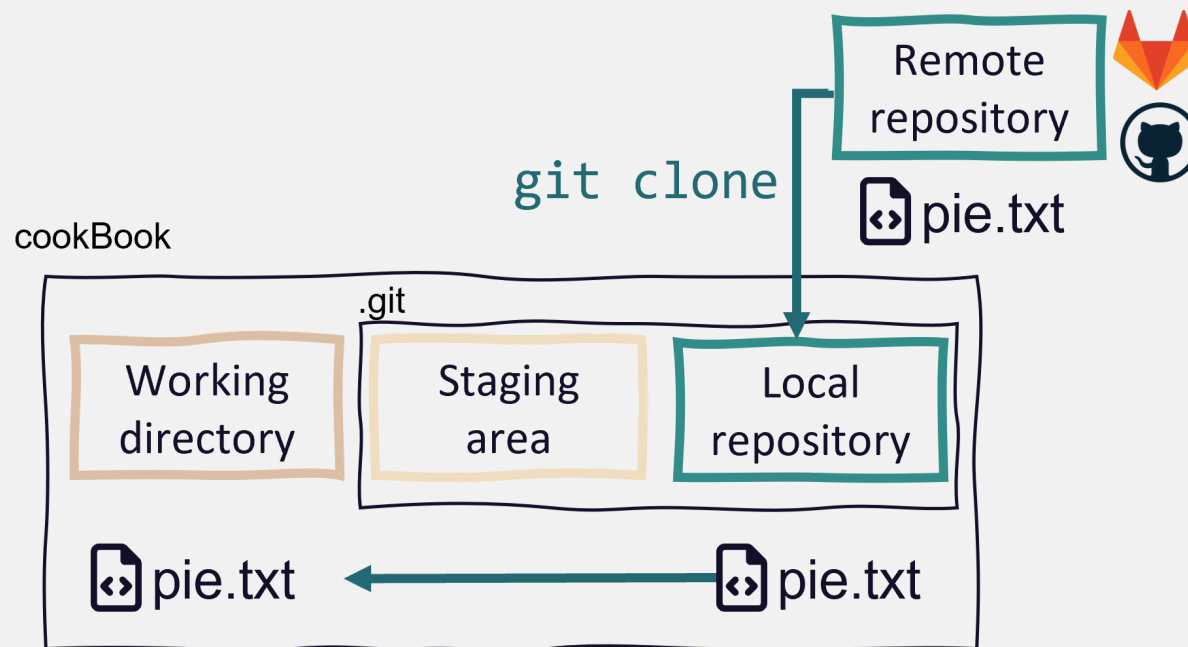
You can clone all public repositories and private repositories if you are a owner/collaborator



Get a repo from a remote

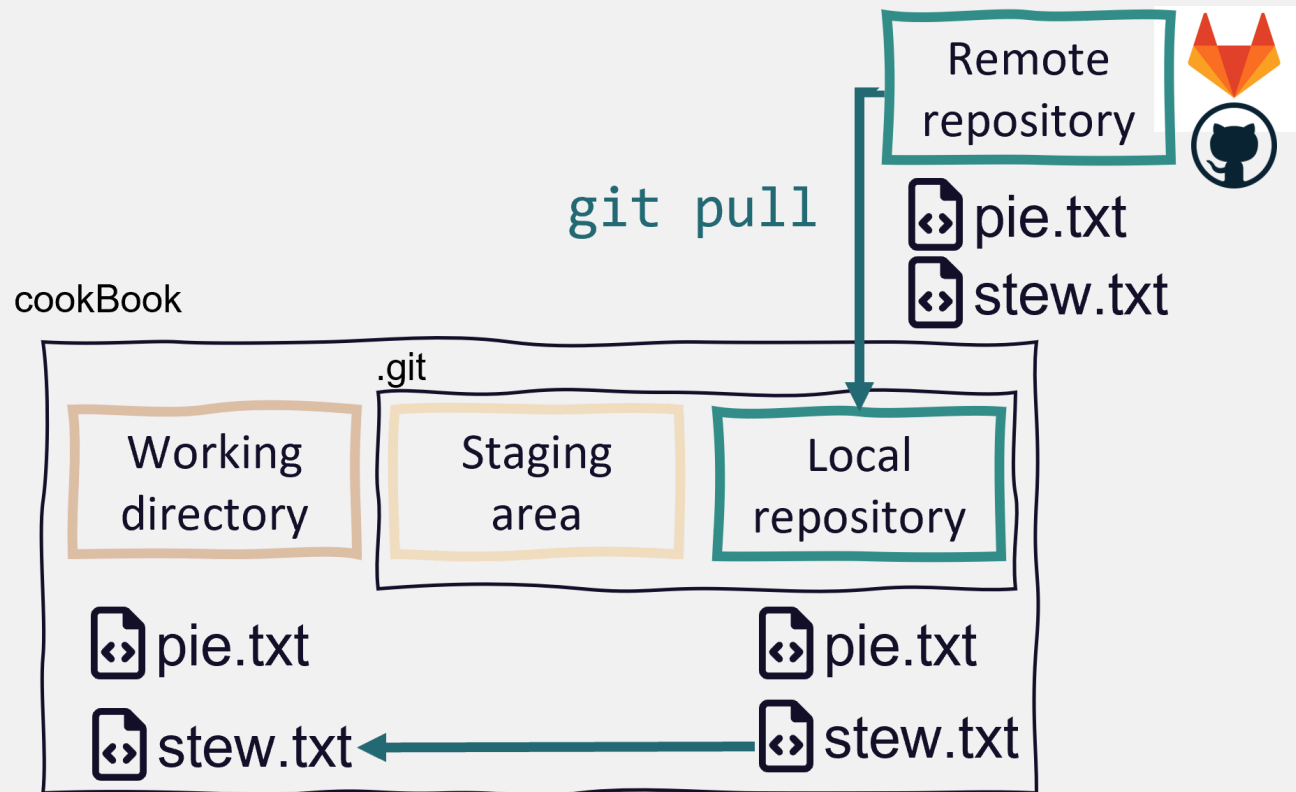
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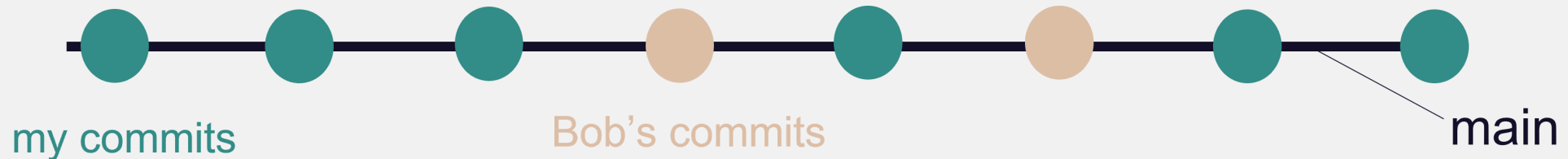


Get changes from the remote

- Local changes, publish to remote: `git push`
- Remote changes, pull to local: `git pull`

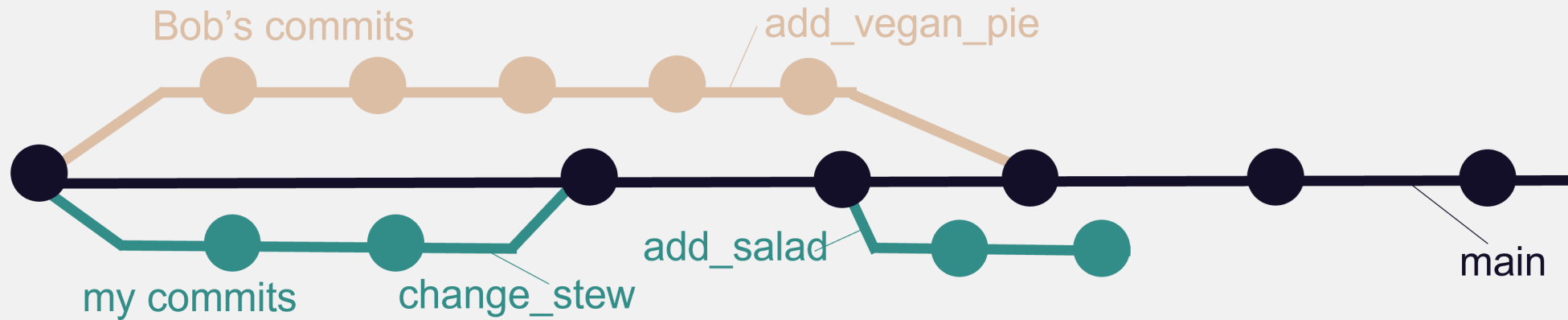


A simple collaboration workflow



- One remote repo on GitHub, multiple local repos
- Idea: Everyone works on the same branch
 - Pull before you start working
 - Push after you finished working

A branching-merging workflow



- One remote repo on GitHub, multiple local repos
- Idea: Everyone works on their **separate branch**
 - **Merge** branch with the main when work is done
- Check out the How-To guide for details

Publishing your work

Remote repositories

- There are **commercial** and **self-hosted** options for your remote repositories
 - Commercial: GitHub, Gitlab, Bitbucket, ...
 - Self-hosted: Gitlab (maybe at your institution?)
- Please be aware of your institutional guidelines
 - Servers outside EU
 - Privacy rules might apply depending on type of data

Public repositories

- Making a repository public is a good way to publish and share your work
- Always add a README.md file
- Always add a LICENSE file
 - This is important to clarify how others can use your work
- Connect your repo with Zenodo to get a DOI

If you are interested, browse some nice GitHub repositories for inspiration (e.g. [Git training repository](#), [Computational notebooks](#), [Repo to publish code from a manuscript](#))

Outlook

- Git can do much more than we covered today
 - Complex collaboration workflows with code review steps
 - Rolling back to previous versions
 - Ignoring files from the repository
 - ...
- GitHub et al. offer many more features
 - Issues, pull requests, code review, project management, ...
 - Host websites, wikis, ...
- Start with the basic workflow and build on that

Next lecture

Summer/Conference break in August and September!

Topic of next lecture t.b.a.

 16th October  4-5 p.m.  Webex

 Subscribe to the mailing list

 For topic suggestions and/or feedback [send me an email](#)

Thanks for your attention

Questions?

Summary of the basic steps

- `git init`: Initialize a git repository
 - Adds a `.git` folder to your working directory
- `git add`: Add files to the staging area
 - This marks the files as being part of the next commit
- `git commit`: Take a snapshot of your current project version
 - Includes time stamp, commit message and information on the person who did the commit
- `git push`: Push new commits to the remote repository
 - Sync your local project version with the remote e.g. on GitHub

Undo things

`git revert`

Revert changes

- Use `git revert` to revert specific commits
- This does not delete the commit, it creates a **new commit** that undoes a **previous commit**
 - It's a safe way to undo committed changes

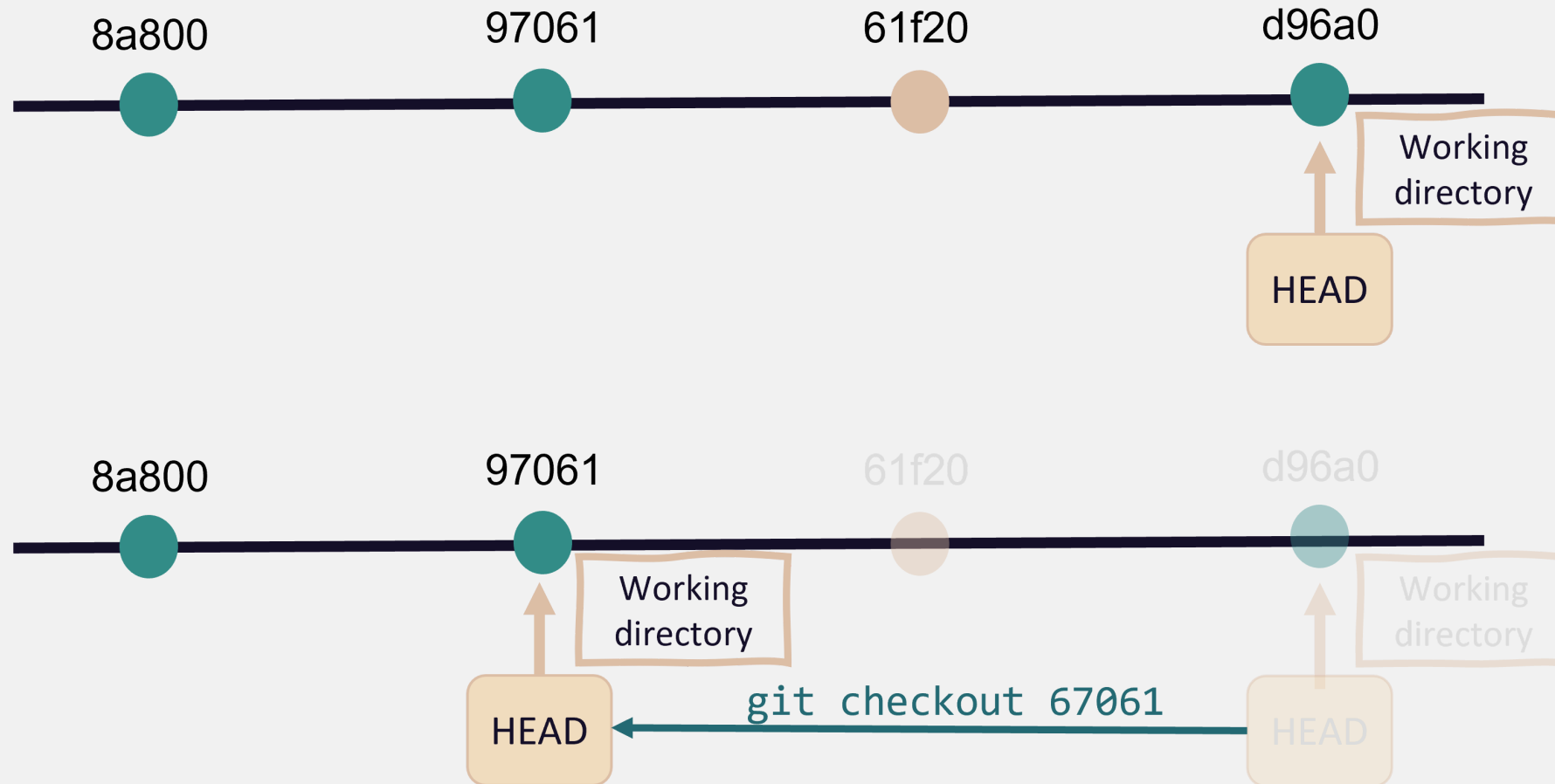


Go back in time

`git checkout`

Checkout a previous commit

Take your work space back in time temporarily with `git checkout`



Ignoring files with `.gitignore`

Ignore files with `.gitignore`

- Useful to ignore e.g.
 - Compiled code and build directories
 - Log files
 - Hidden system files
 - Personal IDE config files
 - ...

Ignore files with `.gitignore`

- Create a file with the name `.gitignore` in working directory
- Add all files and directories you want to ignore to the `.gitignore` file

Example

```
*.html    # ignore all .html files
*.pdf     # ignore all .pdf files

debug.log # ignore the file debug.log

build/    # ignore all files in subdirectory build
```

See [here](#) for more ignore patterns that you can use.